Deep Neural Network Architectures for Phishing Web site Detection and URL Threat Classification

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ABSTRACT

Phishing attacks remain one of the most prominent cybersecurity threats facing modern internet users, causing billions in financial damages through credential theft and identity compromise. Traditional blacklist and heuristic-based defense systems struggle to identify zero-hour phishing domains that dynamically alter their URL structure and host infrastructure. In this paper, we propose CyberGuard-Net, a deep learning pipeline integrating convolutional neural networks and character-level embeddings for real-time phishing detection. Evaluated on a benchmark of 120,000 legitimate and malicious URLs, our model achieves a 99.1% detection accuracy with a 0.2% false positive rate, outperforming commercial intrusion detection systems.

1. INTRODUCTION

The rapid proliferation of digital services and online banking has amplified the threat landscape of social engineering and cybercrime. Among varied attack vectors, phishing continues to be the dominant gateway for enterprise intrusions, ransomware deployment, and credential harvesting. Conventional defense mechanisms rely heavily on static blacklists maintained by security vendors. However, modern adversaries employ automated domain generation algorithms (DGAs), fast-flux DNS configurations, and obfuscated subdomains to circumvent static rules. There is an urgent demand for automated, deep learning-based security systems capable of analyzing structural lexical properties in real-time.

2. PROPOSED METHODOLOGY & SYSTEM ARCHITECTURE

We propose CyberGuard-Net, a multi-stage deep neural network architecture. First, raw URL strings are parsed without relying on third-party DNS lookups. Characters are tokenized into 128-dimensional dense vector embeddings. Next, parallel 1D convolutional layers with multiple filter kernel sizes (3, 5, 7) extract localized n-gram syntax features. A bidirectional LSTM layer models long-range semantic dependencies across query strings, path segments, and top-level domain anomalies. Finally, a fully connected dense layer with dropout regularization (0.4) and sigmoid activation outputs the binary threat probability.

3. EXPERIMENTAL EVALUATION & RESULTS

We conducted extensive experiments on a corpus of 120,000 URLs collected from PhishTank, OpenPhish, and the Alexa Top 1M domains. CyberGuard-Net achieved an overall accuracy of 99.12%, precision of 98.84%, recall of 99.41%, and an F1-score of 99.12%. Comparative baselines, including Random Forest (94.2%) and Support Vector Machines (92.8%), were substantially outperformed. Inference latency averaged 4.2 milliseconds per URL, making it suitable for live browser-level blocking and firewall deployment.

4. CONCLUSION AND FUTURE WORK

We demonstrated that character-level deep neural networks can accurately distinguish malicious phishing domains from legitimate web resources without manual feature engineering or network query overhead. Future work will extend the framework to adversarial evasion detection and cross-lingual phishing detection across internationalized domain names.